

LoQANT User Manual

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Overview

LoQANT (Localization and Quantification of Antigen Nuclear sTaining) is a plugin for ImageJ/Fiji designed to assess the nuclear positivity of monitored antigens in both H-DAB- and fluorescence-stained samples. It is particularly useful in studies where changes in nuclear localization of proteins, such as nuclear receptors and signaling molecules, are of interest. A major advantage of this plugin is that it evaluates the positive signal exclusively within the nucleus, independent of cytoplasmic staining. In addition to identifying nucleus-positive cells, LoQANT can also measure nuclear staining intensity semi-quantitatively or quantitatively, depending on the staining method.

Other advantages of the algorithm include: elimination of inter-observer visual perception bias; minimal supervision required for analysis, thereby reducing the heavy reliance on a trained pathologist or histologist; substantial reduction in analysis time; and a streamlined workflow requiring only a few steps to complete the analysis and measure staining intensity.

General recommendation

When using LoQANT, it is recommended to follow good image analysis practice to obtain valuable results, such as fixation, staining, image acquisition settings, exposure, contrast and brightness being consistent across a cohort of samples (Cromey, 2010). For LoQANT to work properly, you need to specify the size of the nuclei in pixels. This will filter out artefacts or overlapping cores. It is also advisable to test which of the automatic thresholding algorithms are suitable for the measured samples.

Cromey DW. Avoiding twisted pixels: ethical guidelines for the appropriate use and manipulation of scientific digital images. *Sci Eng Ethics*. 2010;16(4):639-67.

1 Installation

1.1 Requirements

- **Operating system:** LoQANT works on Windows, Mac and Linux operating systems (tested on Windows 10, Windows 11, macOS Monterey 12.5., and Ubuntu 22.04 LTS operating systems)
- **Software:** The latest version of FIJI (last tested on version 2.14.0/1.54f)
- **Image formats:** the image format depends on type of method:
 - H-DAB: one RGB image, white background
 - Fluorescence: two RGB or grayscale images of separate channels, black background
 - Preprocessed images: H-DAB: two grayscale images of separate channels, white background

1.2 Download Fiji

Download the latest version of FIJI at <https://imagej.net/software/fiji/downloads> according to your operating system and install it.

1.3 Download LoQANT

LoQANT is freely available at <https://github.com/Keri-histo/LoQANT>.

1. Download the LoQANT_.py and LoQANT_batch.py files.
2. Find the Fiji.app folder on your computer.
3. Open the Fiji.app folder and place LoQANT_.py and LoQANT_batch.py files to folder called „plugins“.
4. Launch Fiji and LoQANT and LoQANT_batch are now to be found in Plugins menu. (Note: If the Fiji has been already open, it needs to be restarted.)

2 Running LoQANT

1. Open Fiji and open image or images for analysis. The image (images) must be open before launching LoQANT.
2. In the plugins menu, select LoQANT. You will find it near the bottom of the list. This will open the initial dialog box (see below).

Setting parameters for analysis

Choose relevant staining method:

Method:

Image requirements:

- * H-DAB: one RGB image (white background)
- * Fluorescence: two RGB or grayscale images of separate channels (black background)
- * Preprocessed images: H-DAB: two grayscale images of separate channels (white background)

Size of nuclei:

Set size of nuclei is highly recommended.

If there is no input, the range 0 - Infinity will be applied.

Set size of nuclei (px unit):

Minimum:

Maximum:

Nuclear intensity measurement:

Additional settings:

Nuclear segmentation:

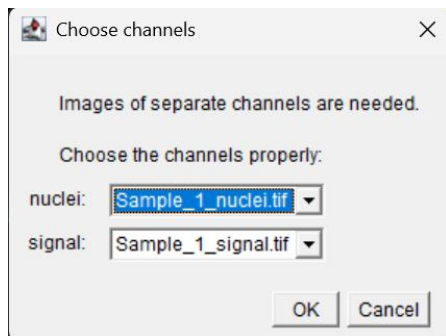
Signal thresholding:

Positive signal in nuclear area (%):

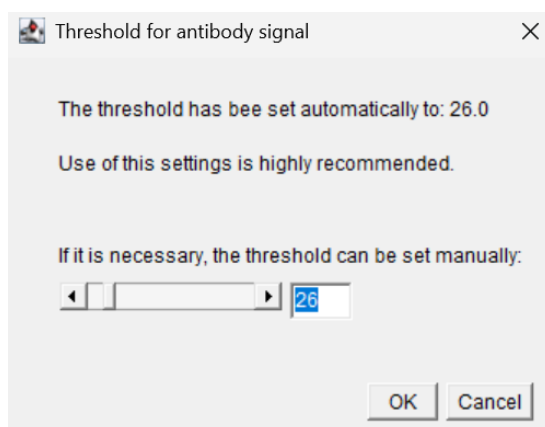
OK Cancel

3. Choose the staining method used for your sample from the menu. You can select from three options: „H-DAB“, „fluorescence“ or „preprocessed images: H-DAB“.
4. Specify the range of nuclei sizes (in pixel units) to be included in the analysis. The size specification is used to filter out particles that are too small (which are usually artefacts) and also clusters of cells where the analysis is distorted by the overlay. If you do not specify the nuclei size, the default setting of „0 – Infinity“ will be applied.
5. Choose if you want to measure the intensity of the nuclear staining. By default, no measurement is set („do not measure“ option). In this case, LoQANT will only analyse the percentage of positive cells. There are three intensity measurement options to choose from:
 - a. A semiquantitative measurement, which is suitable for DAB-stained samples (choose the „semiquantitative (rec for DAB)“ option). This method measures the number and percentage of the positive nuclei in the following categories: weak, medium, and high intensity. It then calculates the histoscore and returns the overall positivity category for a given field of view.

- b. A quantitative measurement of intensity value for each positive nuclei („quantitative: positive nuclei (rec for fluorescence)“ option). This method measures the intensity of staining as the mean gray value in nuclei considered as positive. Quantitative measurement is recommended for fluorescent samples.
 - c. The quantitative measurement of all detected nuclei („quantitative: all nuclei (rec for fluorescence)“ option). It measures intensity of staining as the mean gray value in all detected nuclei, regardless of whether they are considered positive or not. Quantitative measurement is recommended for fluorescent samples.
6. In the Additional Settings section, you can select the algorithm for segmenting the nuclei and the positive signal. You can use all of the AutoThreshold algorithms available in Fiji (Default, Huang, Huang2, Intermodes, IsoData, Li, MaxEntropy, Mean, MinError, Minimum, Moments, Otsu, Percentile, RenyiEntropy, Shanbhag, Triangle and Yen). You can also set the percentage of nuclei that must have a positive antibody signal for all nuclei to be considered positive. Default value is 80 %. It is set based on human evaluation of testing sample set, however it can be changed.
7. If you select either the „fluorescence“ or „preprocessed images“ method, the next dialog box will ask you to specify which of the opened images is for nuclei and antibody signal. All images opened in Fiji will be listed in the menu.



8. The nuclei and antibody signal will be segmented automatically using AutoThreshold algorithms that you selected in initial dialog.
9. If the antibody thresholding does not work as expected, you can change the threshold value using the slider. The automatically set threshold value will be displayed in the upper right corner of the dialog box. After confirming the threshold value, the plugin will evaluate the image (see below).



IMPORTANT NOTES:

- 1. The ROI Manager must stay open during whole the analysis.**
- 2. Incorporation of AI nuclei segmentation of H-DAB images to the LoQANT workflow:**
If using external tools in Fiji for nuclei segmentation such as StarDist, Trainable WEKA, etc. followed with “preprocessed images: H-DAB”, be sure that the generated mask of nuclei has white background (value 255 in 8-bit image) as .
- 3. Incorporation of AI nuclei segmentation or preprocessing on fluorescent images to the LoQANT workflow:**
If the fluorescent images need to be preprocessed or external tools in Fiji for nuclei segmentation as StarDist, Trainable WEKA, etc. are used, be sure that the generated images of mask of nuclei has black background (value 0 in 8-bit image) as image of signal. Then the “fluorescence” method should be used.

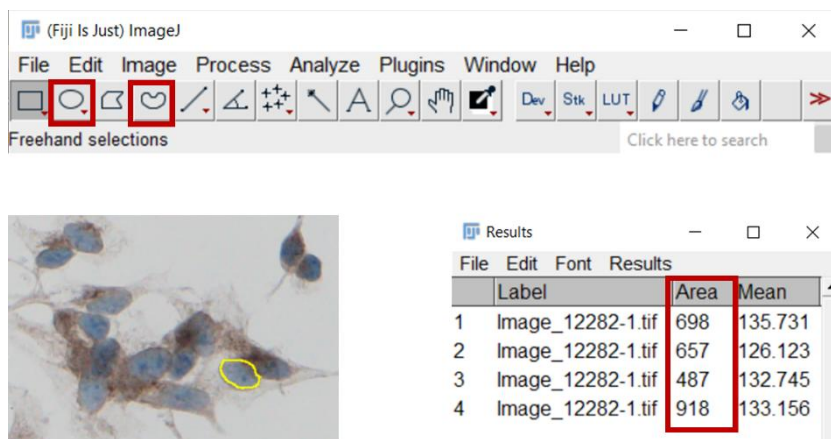
3 Running LoQANT_batch

1. Prepare folder with images to be analysed (“H-DAB”) or folder with nuclei and folder with signal (“fluorescence” or “preprocessed images”). No image needs to be opened in Fiji in advance.
2. In Fiji Plugins menu choose LoQANT_batch. You will find it near the bottom of the list. This will open the initial dialog box (see Section 2, step 2).
3. The first dialog is the same as in basic analysis. Set parameters for analysis of set of samples: method, specify nuclei size, intensity measurement, algorithms and % of positive signal in nuclei (see Section 2, steps 3 – 6).
4. The next dialog will ask where the folder of images for analysis is located.
 - a. If you select “H-DAB” method, the dialog will ask for one folder of images.
 - b. If you select “fluorescence” or “preprocessed images” method, the first dialog will ask for the folder of images of nuclei and the following for the folder of signal.
5. For batch processing, AutoThreshold will be applied for antidoby signal.
6. The results are written in .csv file and saved to the folder with images (“H-DAB”) or to the folder with images of signal (“fluorescence” or “preprocessed images”).
7. Important note for viewing .csv files in Excel: Local specifics may cause incorrect display of results when opening .csv file in Excel. LoQANT_batch use semicolon as a separator in .csv file (used in for example in many European countries). When you use Excel with different separator, for example comma (US, UK, Canada, etc.) please open .csv via Excel’s “Import Wizard” and select the delimiter.

4 Input parameters determination

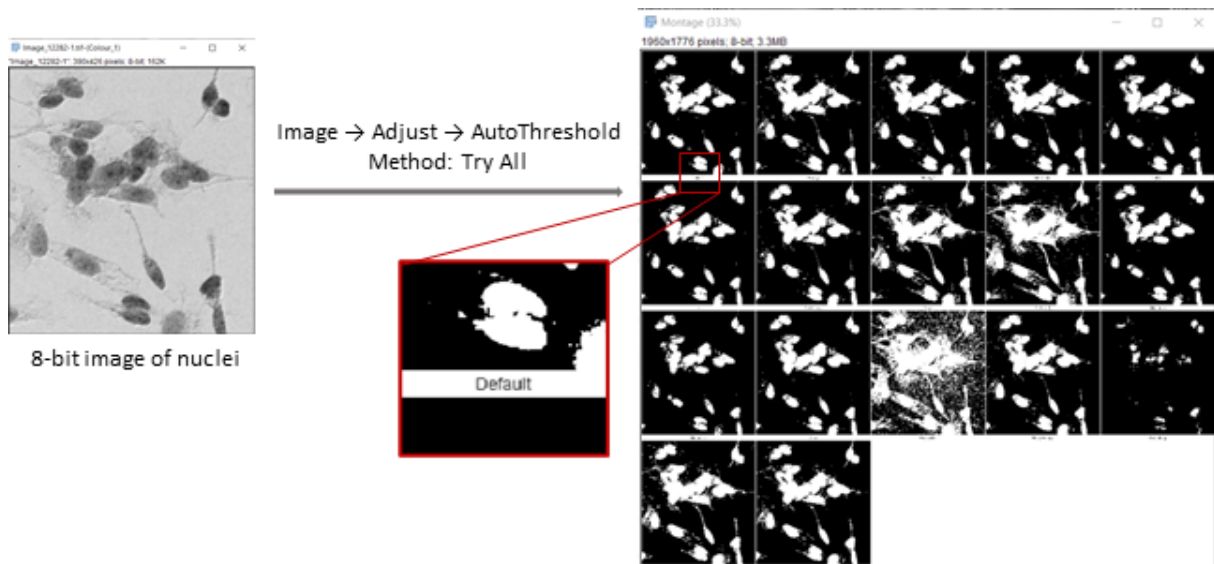
4.1 How to measure the size of a nucleus

1. LoQANT requires the size in pixel units. So, before you start, check units of measurement: select "Set Scale" in the "Analyze" menu and check in the corresponding dialog box. If you've set any units, click "Click to Remove Scale" to remove them.
2. To measure the size of the nuclei in the analysed sample, select either "oval" or "freehand selection" in the toolbar and mark the nucleus. You can then measure its size either via the "Analyze - Measure" menu or by pressing the "m" key. The Result Table will open, the size is in the Area column.



4.2 Selecting the AutoThreshold algorithm

1. You can use all the algorithms for automatic thresholding available in AutoThreshold (ver. 1.18.0), which is part of Fiji.
2. To select the appropriate algorithm for nuclei segmentation, you will need an 8-bit image representing the nuclei.
3. If the staining is fluorescence, the channel is separate and just needs to be converted to 8-bit (if it is not already) via the "Image - Type - 8-bit" menu.
4. If the staining is DAB, you need to deconvolve the colours first. This is done via the menu "Image - Color - Color Deconvolution". After deconvolution the image is 8-bit.
5. Navigate menu "Image - Adjust - AutoThreshold" and select "Method: Try all". It is also necessary to check whether the objects are white objects on a black background (in the case of fluorescence) or not (in the case of DAB). After confirmation, the segmentation result is displayed using all 14 available algorithms. The name of the algorithm used is below each image. Choose the one most suitable for your sample.



5 Test images

You can test LoQANT function using provided test images.

H-DAB.tif

- RGB image
- open this image in Fiji
- select method: H-DAB
- set nuclei size to: 300-1500

fluorescence_nuclei.tif and fluorescence_signal.tif

- RGB images of separate fluorescence channels
- open both images in Fiji
- select method: fluorescence
- set nuclei size to: 100-800
- in following dialog box, set:
 - fluorescence_nuclei as image represents nuclei
 - fluorescence_signal as image represents signal

preprocessed_haematoxylin_nuclei.tif and preprocessed_DAB_signal.tif

- 8-bit grayscale images generated as result of color deconvolution of H-DAB image
- open both images in Fiji
- select method: preprocessed H-DAB
- set nuclei size to: 300-1500
- in following dialog box, set:
 - preprocessed_haematoxylin_nuclei as image represents nuclei
 - preprocessed_DAB_signal as image represents signal